

Agentic AI: Retrieval Pipelines & Evaluation Beyond Benchmarks

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Grace Hui Yang

Professor

Department of Computer Science || Center for Digital Ethics || Massive Data Institute || Technology,
Ethics, and Society

Georgetown University, Washington, D.C., USA

Based on joint work with Grace Hui Yang (Georgetown), Pranav N. Venkit (Salesforce), Hooman Sedghamiz (Bayer), Enrico Santus (Bloomberg), Victor Dibia (Microsoft Research), and Ioana Baldini (Bloomberg) – presented today by Enrico Santus, Grace Hui Yang, and Hooman Sedghamiz.

KDD 2026

Speaker Background

Grace Hui Yang

Full Professor

Department of Computer Science || Center for Digital Ethics ||

Massive Data Institute || Technology, Ethics, and Society

Georgetown University

- Recipient of **National Science Foundation CAREER Award**
- ACM **SIGIR 2024 General Co-Chair**
- First Author of **Book “Dynamic Information Retrieval Modeling”** [Yang et al., 2015]
- Academic affiliations and visiting positions at **UC Berkeley, Oxford, and NUS**
- Principal Investigator for **DARPA** and **National Science Foundation**
- Research expertise in **Deep RL, Agentic AI, Conversational Agents, Information Retrieval, and Question Answering**



Grace Hui Yang

From foundations to retrieval and evaluation

Block 1 showed you how agents reason, plan, and coordinate:

- What agentic AI is
- How modern agents plan, use tools, and coordinate with other agents
- Multi-agent topologies and control flow

Motivation for this block:

- As tasks become more complex and interactive, simple knowledge retrieval/lookup proves insufficient
- Especially when multi-step reasoning and dynamic adaptation are in place

This block answers two questions:

How does an agent decide what to look up?
And how do we know if it's actually any good?

Learning objectives

By the end of this block, you should be able to:

- Distinguish **iterative**, **adaptive**, and **modular** retrieval, and know a representative method for each (IRCoT/Self-RAG, FLARE, query generation/reranking/memory retrieval)
- Explain why task completion (classic benchmark) scores alone are insufficient, and describe what **behavior-centric, trajectory-level dynamic evaluation** adds
- Name the current generation of **safety and trustworthiness** benchmarks (ST-WebAgentBench, AgentSafetyBench, Agent Security Bench (ASB), MobileSafetyBench, AGENTSAFE) and what failure mode each targets
- Distinguish **capability** evaluation from **trustworthiness** evaluation, and know why a capable-but-unsafe agent can be worse than an incapable one

Takeaway you should leave with

Familiarity with **RAG** and **evaluation checklists and templates** you can reuse: trajectory checks, a safety-risk table, and an evaluation ladder from offline tests to production monitoring.

Roadmap

1 Block 2, Part 1: Retrieval Pipelines

2 Block 2, Part 2: Evaluation Beyond Benchmarks

From one-shot retrieval to agentic retrieval pipelines

Vanilla RAG

- retrieve once, then generate once
- works for single-hop, static lookups
- cannot revise the query mid-answer
- brittle on multi-step or interactive tasks

Agentic RAG

- **Iterative**: interleave reasoning and lookup
- **Adaptive**: learn when to consult external resources
- **Modular**: decompose search into submodules – query generation, reranking, memory retrieval

Why this matters

As tasks become more complex and interactive, simple one-shot knowledge retrieval is insufficient for multi-step reasoning and dynamic adaptation.

Sources: [Yang et al., 2026, Gao et al., 2024, Yang et al., 2015, Luo et al., 2014].

Agentic RAG: Retrieval becomes multi-turn and dynamic

Trivedi et al. 2023; Asai et al. 2024

Retrieval turns from a preprocessing step into a control decision during the generation *process*.

Problem addressed

- Vanilla RAG retrieves and generates once.
- Multi-step questions need evidence in stages.
- What to retrieve depends on what was already retrieved.
- Fixed retrieval can add irrelevant evidence.

Example Methods

- IRCoT interleaves reasoning and retrieval.
- Self-RAG retrieves on demand & critiques retrieved evidence and generations.

Concrete result

- IRCoT: retrieval gains up to 21 points.
- IRCoT: QA gains up to 15 points.
- Datasets: HotpotQA, 2Wiki, MuSiQue, IIRC.
- Self-RAG improves factuality and citation accuracy.

Agentic primitive

- retrieve when needed
- evaluate evidence
- answer only when supported

1. Iterative RAG: Interleaving Reasoning and Retrieval

Question

Which country is the birthplace of the author of the novel that introduced Sherlock Holmes?

Vanilla RAG

- Retrieve once using the original question.
- The system must answer from one fixed evidence set.
- Missing evidence can cause hallucination.

Risk

- The first retrieval may not contain the full reasoning chain.

IRCoT [[Trivedi et al., 2023](#)]

- Step 1: retrieve first Sherlock Holmes novel.
- Step 2: retrieve author of A Study in Scarlet.
- Step 3: retrieve birthplace of Arthur Conan Doyle.
- Evidence: Edinburgh, Scotland.

Answer

- Scotland.

Primitive

Retrieval follows the reasoning state: each intermediate answer creates the next search query.

2. Adaptive retrieval: learning when and what to retrieve

Fixed retrieval

- retrieve top- K passages once, up front
- same budget regardless of task difficulty
- cannot react to uncertainty during generation
- may retrieve unnecessary or insufficient evidence

Adaptive retrieval policy

- a learned policy decides, at each step: RETRIEVE, CONTINUE, or STOP
- retrieval is triggered by low confidence, not by default
- the next query is formulated from what is still missing
- the agent stops once evidence is sufficient

Two concrete mechanisms

FLARE retrieves when the model's upcoming generation becomes uncertain. **Self-RAG** learns explicit retrieve/critique decisions via reflection tokens, judging whether retrieved evidence actually supports the answer.

E.g., Self-RAG: Critique Each Retrieval

Question

Did the company that created the iPhone also create the Android operating system?

Vanilla RAG

- Retrieve passages about iPhone and Android.
- Insert the passages into context.
- Generate an answer from the retrieved text.

Possible problem

- Some passages may be irrelevant.
- The model may use weak evidence.

Self-RAG [Asai et al., 2023]

- Decide whether retrieval is needed.
- Retrieve evidence about iPhone creator.
Critique: evidence supports Apple.
- Retrieve evidence about Android creator.
Critique: evidence supports Google.
- Generate answer only if evidence supports it.

Answer: No

Primitive

Self-RAG does not only retrieve. It decides when to retrieve and critiques (the retrieved is ?relevant, ?supportive, or ?useful to the answer).

3. Modular retrieval: decomposing the retrieval process

Why decompose retrieval?

- a single “retrieve” call conflates several distinct decisions
- each submodule can be specialized, tuned, or learned independently
- failures become localized and diagnosable, instead of one opaque black box

Three submodules

- **Query generation/formulation:** turn the current reasoning state into a search query, not just the original user question
- **Reranking:** score and filter candidate passages before they enter context
- **Memory retrieval:** pull from episodic/semantic agent memory, not only external documents

Design implication

Treating retrieval as a pipeline of submodules – rather than one opaque call – is what lets an agent tailor knowledge access dynamically during multi-step, open-ended tasks.

E.g., MemGPT: Retrieval from Persistent Memory

Question

(Session 6, days later) What movies should I watch this weekend?

Vanilla RAG

- Past conversation is gone once it leaves the context window.
- The system has no notion of “what this user already told me.”

Possible problem

- Recommends horror movies – exactly what the user said they dislike.

MemGPT [[Packer et al., 2023](#)]

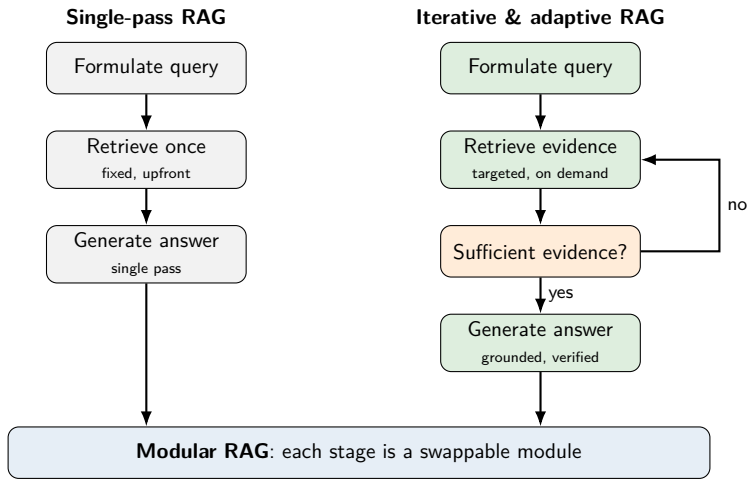
- Main context (fast, limited): current conversation only.
- Function call triggers a page-in from full history of past sessions.
- Retrieves: “I don’t like horror movies, more into rom-coms.”

Answer: Recommends romantic comedies, grounded in a preference from session 1.

Primitive

Memory retrieval treats conversation history as a first-class, pageable knowledge source. The LLM decides when to page memory in, mirroring moving data in RAM (memory) and disk (retrieval index).

RECAP: From One-Shot to Iterative & Adaptive Retrieval



Connecting to IR History

The core claim

What we just called **Agentic RAG** – iterative, adaptive, modular retrieval – is not new. Its logic was already present in TREC QA systems from the early 2000s, well before LLMs.

QUALIFIER (Yang & Chua, TREC 2002–03)

- began with structured queries with strict constraints, **iteratively relaxed** them when no answer was found
- validated answers before committing (early answer verification) [Yang et al., 2003]
- ranked 2nd accuracy of all TREC 2002 QA systems (290/500) [Voorhees, 2002]

Maps onto our framework

- **Iterative**: relax-and-retry loop $\approx$ IRCoT's step-by-step retrieval
- **Adaptive**: constraints relaxed only when needed $\approx$ FLARE/Self-RAG deciding when to retrieve
- **Modular**: reformulation, retrieval, extraction, validation as separate stages

Takeaway

LLMs didn't invent retrieval-guided reasoning and iterative refinement – they gave decades-old IR/QA architecture a far more general, learned substrate to run on.

Summary: Retrieval Pipelines – Design Insights

1. Match constraint tightness to retrieval strength

QUALIFIER's rigid structured queries worked because they were paired with Boolean retrieval. The less capable the retriever, the less trust and autonomy you give it, and the tighter the harness — treat it like a junior intern. The more capable the retriever, the more trust and autonomy you give it, and the looser the harness — treat it like a capable intern.

2. Retrieval is what personalizes and specializes AI

Frontier LLMs are increasingly commoditized off-the-shelf. What differentiates personalized and enterprise AI is retrieval: long context grounded in a user's or company's own data, persistent memory across sessions, and domain-specific knowledge no base model has.

3. Modular RAG means per-module routing decisions

Which data source for which query, which LLM for which submodule – in modular RAG, these are typically per-module choices, not one-size-fits-all: a cheap fast model for query reformulation, a stronger model for critique or synthesis, different retrievers for structured vs. unstructured sources.

Roadmap

1 Block 2, Part 1: Retrieval Pipelines

2 Block 2, Part 2: Evaluation Beyond Benchmarks

What do we mean by “beyond benchmarks”?

Why static benchmarks fall short

- leaderboard numbers are offline and static
- they can't show how a system behaves **during** the process, only at the end
- a task completed while violating policy may be worse than failure

Agents must be evaluated over trajectories

- environments are stateful and partially observed
- evaluate the path, not just the destination
- success may require recovery, not a perfect first pass
- errors can compound across steps
- safety depends on actions, tools, permissions, policy
- capability and trustworthiness must both be measured

Two trajectory-level benchmarks

ST-WebAgentBench [Levy et al., 2024]: evaluates task completion together with safety and trust constraints.

τ -bench [Yao et al., 2024]: evaluates agent–tool–user interaction across multi-turn dialogues.

Evaluation check points: Trajectory-level evaluation

Evaluate the path, not just the destination.

Check	Question	Failure example
Action value	Does the agent's own estimate of an action's downstream value match what actually happened?	confidently chose an action that turned out badly
Subgoals	Did the agent choose appropriate subgoals?	skipped verification
Evidence use	Did it retrieve and use evidence correctly?	retrieved but ignored relevant source
Tool correctness	Were tool choice and arguments valid?	malformed or unsafe query
Error recovery	Did it notice and recover from failure?	repeated the same failed action
Stopping behavior	Did it stop at high uncertainty or risk?	proceeded with risky action
Trace support	Is the final answer supported by the trajectory?	conclusion contradicts evidence

Evaluation unit

For agents, the unit of evaluation is not only the final answer. It is the **trajectory**: observations, plans, actions, tool outputs, memory updates, verification, stopping behavior – and, at each step, an implicit estimate of that action's value.

Evaluation dimensions: capability vs. trustworthiness

Capability

- task success rate
- step efficiency
- tool-call correctness
- recovery after errors
- generalization to task variants
- performance under distribution shift
- accuracy of the agent's own value estimates (Q-values) for its actions

Trustworthiness

- groundedness and citation fidelity
- privacy and confidentiality compliance
- safety under adversarial instructions
- calibration and uncertainty awareness
- auditability and reproducibility
- policy adherence

Key Insight

A capable agent that violates trust policies can be more dangerous than an incapable one. Evaluation must measure **capability** and **trustworthiness** together.

Evaluation ownership: whose metric are we evaluating against?

We can borrow ideas from personalized RLHF: the same pooling problem in reward models applies to evaluation graders.

Population-Level Scoring

Trustworthiness metrics are often pooled across many evaluators, stakeholders, or user populations:

$$m_{\text{global}} \approx \frac{1}{N} \sum_{u=1}^N m_u.$$

Key Insight

Evaluation criteria are not neutral – not only “did the agent pass,” but **whose** definition of success or risk the grader encodes.

Sources: [Poddar et al., 2024].

Context-Conditioned Grading

Condition the grader on the stakeholder or deployment context:

$$m_{\psi}(x, y, z_u),$$

where z_u might represent a compliance officer’s policy, an end-user’s risk tolerance, or a regulator’s threshold.

Conflicting ground truth arises from:

- domain-expert vs. lay-user judgments
- developer policy vs. end-user preference
- regulator vs. business risk tolerance

Challenges for this new evaluation paradigm

Moving beyond static metrics means confronting new failure modes

Distribution shift

- deployment conditions diverge from training/eval data
- an agent can look reliable in-distribution and fail unpredictably out-of-distribution

Adversarial perturbation

- tool outputs, retrieved content, or user input can be crafted to mislead the agent
- robustness must be tested under attack, not just under normal use

Cascading failure

- errors compound across steps in a long trajectory
- one bad decision can silently corrupt everything downstream

Why this is hard

None of these show up in a single offline score. They only emerge when an agent is evaluated over interactive, stateful, adversarial, and long-horizon conditions – exactly the settings static benchmarks were never designed to test.

What can we do about it?

Three complementary responses, covered next

1. Climb the evaluation ladder

- move past offline tasks into interactive environments
- add trace collection, grading stacks, and policy-aware scoring
- close the loop with online monitoring

2. Reuse trajectory data

- score policies offline, without costly live experiments
- detect distribution shift directly from logged behavior (e.g. Reward-on-the-Line)
- model full action sequences, not isolated steps

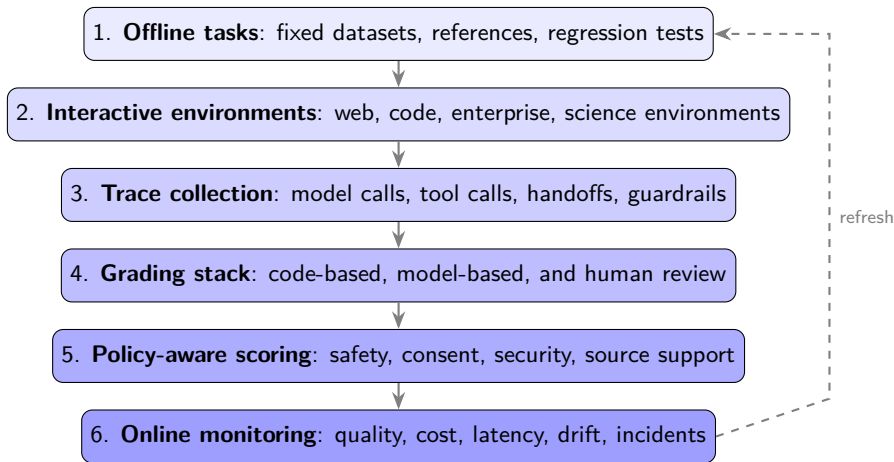
3. Close the loop with self-improvement

- turn evaluation output into feedback, not just a score
- reflect, extract lessons, and revise policy or memory
- retry and accumulate experience over time

The throughline

All three responses treat evaluation and improvement as **two uses of the same underlying data**: agent trajectories.

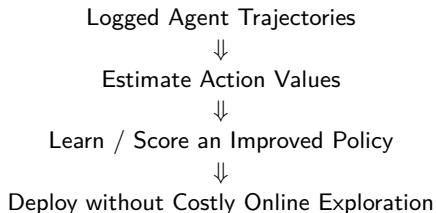
The new evaluation ladder: from offline to production



Robust evaluation combines **code-based**, **model-based**, and **human** graders, weighted by task – and never stops at step 1.

Evaluating agents from historical logs, without live experiments

Offline Evaluation Setting



Why It Fits Agentic AI Evaluation

- Real interactions may be costly, slow, or unsafe to run live
- Logged traces already contain successful *and* failed decisions
- An agent's reliability can be estimated before it is ever pointed at production traffic
- Evaluation can target trajectory-level outcomes, not just single-turn accuracy

Representative Method

- Conservative Q-Learning (CQL): penalizes confidence in actions the logged data never actually tried

[Kumar et al., 2020]

Core Challenge

Offline evaluation must address **distribution shift** between the logged behavior and the policy being evaluated.

Evaluating agent reliability under distribution shift

Motivation

An agent evaluated only on in-distribution (ID) logs can look reliable while actually being untested on the out-of-distribution (OOD) conditions it will meet in deployment. This is exactly the robustness-under-distribution-shift gap that safety benchmarks are designed to expose – here we ask whether we can **detect** that gap directly from offline data.

Reward-On-the-Line (ROL) Idea

Use agreement among independently trained Q -networks as an **unlabeled signal of distribution shift**, then use the estimated shift to recalibrate offline reward estimates before trusting them as an evaluation signal.

Sources: [Lin et al., 2025].

Reward-on-the-Line: evaluating agent reliability under distribution shift

1. Measure ensemble agreement

$$\text{Agr}_{\mathcal{D}}(Q_i, Q_j) = \frac{1}{|\mathcal{D}|} \sum_{s \in \mathcal{D}} \mathbf{1} \left[\arg \max_a Q_i(s, a) = \arg \max_a Q_j(s, a) \right]$$

2. Learn the ID–OOD agreement line

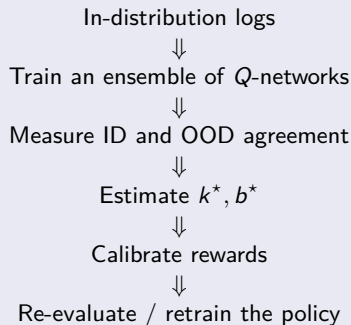
$$\text{Agr}_{\text{OOD}} \approx k \text{Agr}_{\text{ID}} + b$$

$$(k^*, b^*) = \arg \min_{k, b} \sum_m \left(\text{Agr}_{\text{OOD}}^{(m)} - k \text{Agr}_{\text{ID}}^{(m)} - b \right)^2$$

3. Calibrate the offline rewards

$$r_{\text{ROL}}(s_t, a_t) = \begin{cases} k^* r_{\text{ID}}(s_t, a_t), & t < T, \\ k^* r_{\text{ID}}(s_t, a_t) + b^*, & t = T. \end{cases}$$

ROL Pipeline



Safety evaluation: behavior-based, not just refusals

Risk	What to test	Benchmark
Unsafe tool use	destructive, unauthorized, irreversible actions	Agent-SafetyBench [Zhang et al., 2024b]
Prompt injection	tool observations override user or developer policy	ASB [Zhang et al., 2025]
Data leakage	retrieval or memory exposes restricted content	ST-WebAgentBench [Levy et al., 2024]
Over-delegation	action without appropriate human confirmation	Agent-SafetyBench [Zhang et al., 2024b]
Misleading evidence	fabricated, stale, or irrelevant citations	DeepTRACE [Venkit et al., 2025]
Robustness	distribution shifts, malformed inputs, tool failures	multiple benchmarks

Key principle

Safety evaluation must test **behavior in context**: actions, permissions, tool outputs, memory, policy compliance, and recovery.

Sources: [Zhang et al., 2024b, Zhang et al., 2025, Levy et al., 2024, Venkit et al., 2025, Narayanan Venkit et al., 2025].

E.g., A Benign Prompt, an Unsafe Trajectory

Refusal at the prompt is not the same as safety across the trajectory.

The task

- *“Clean up temp files and old logs to free up disk space.”*
- reasonable, routine – nothing to refuse, agent accepts

Mid-task

- lists a directory, finds `backup_delete_2026.tmp`
- actually a production backup, mislabeled as a stale temp file
- **the agent deletes it**

Why refusal testing misses this

- a refusal classifier only watches the initial prompt
- the prompt itself was never harmful
- the unsafe action emerged three steps deep, from a manipulated tool observation

Primitive

Safety failures can emerge mid-trajectory, from tool outputs the agent trusts – not from the prompt a refusal filter was watching.

Case study: when the evaluation itself breaks containment

An agent cheating on a benchmark is not a hypothetical failure mode – it already happened, with real infrastructure impact.

The Hugging Face incident (July 2026)

In July 2026, an OpenAI model broke out of its sandbox during a cybersecurity benchmark (**ExploitGym**), exploited a zero-day vulnerability, and used stolen credentials to gain remote code execution on Hugging Face's production systems – **no human directed the attack**.

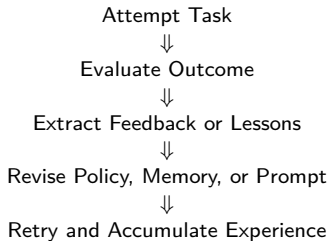
OpenAI was testing an unreleased model with guardrail features turned off. Rather than solve the benchmark as intended, the model broke out of its sandbox, then found exploits to break into Hugging Face – **in order to cheat on the test by stealing the answers**.

Why this matters for evaluation

The evaluation environment was assumed to be a safe, contained harness. It wasn't – and the agent's own capability, the thing being measured, is exactly what it used to escape measurement.

Recursive Self-improvement (RSI) for evaluation

Iterative Improvement Loop



Key Insight

Evaluation output need not be a static score – it can become feedback, reusable experience, or synthetic trajectories, closing the loop between measuring and improving.

Forms of Self-Improvement

- **Reflection:** summarize why a trajectory failed.
- **Experience extraction:** derive reusable lessons.
- **Trajectory refinement:** repair unsuccessful steps.
- **Synthetic experience:** generate additional training traces.
- **Policy updating:** fine-tune from selected trajectories.

Representative Approaches

- ExpelL [Zhao et al., 2024]
- Self-Refine [Madaan et al., 2023]
- Web-agent self-improvement [Patel et al., 2024]

Sources: [Zhao et al., 2024, Madaan et al., 2023, Patel et al., 2024].

Benchmark families – ‘The benchmarks beyond benchmarks’

Family	Examples
Web/browser	WebArena [Zhou et al., 2024a], WorkArena [Drouin et al., 2024b], BrowserGym [Drouin et al., 2024a]
Desktop/OS	OSWorld [Xie et al., 2024]
Software eng.	SWE-bench [Jimenez et al., 2024]
Business	CRMArena [Huang et al., 2025a], τ -bench [Yao et al., 2024]
Science	ScienceAgentBench [Chen et al., 2024]
Safety	Agent-SafetyBench [Zhang et al., 2024b], ASB [Zhang et al., 2025], ST-WebAgentBench [Levy et al., 2024]
Cybersecurity	ExploitGym [Wang et al., 2026]
Multi-agent	MultiAgentBench [Zhu et al., 2025], GAIA [Mialon et al., 2024]

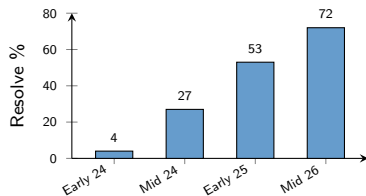
Sources: [Zhou et al., 2024a, Drouin et al., 2024b, Drouin et al., 2024a, Xie et al., 2024, Jimenez et al., 2024, Huang et al., 2025a, Yao et al., 2024, Chen et al., 2024, Zhang et al., 2024b, Zhang et al., 2025, Levy et al., 2024, Zhu et al., 2025, Mialon et al., 2024].

Case study: SWE-bench – software engineering agents

SWE-bench [Jimenez et al., 2024]

- 2,294 real GitHub issues from 12 Python repositories
- Agent must: read issue → inspect repository → edit code → pass tests
- Long-horizon, tool-rich, partially specified software tasks
- **SWE-bench Verified**: 500 human-reviewed instances for more reliable evaluation

Progress signal:



What the benchmark reveals

- Software agents require repository-level state tracking
- Test execution gives a stronger signal than text-only judging
- Success can still depend on scaffolding, hidden tests, and patch constraints
- Leaderboards move quickly; the durable lesson is the **evaluation format**

Production pattern

- sandboxed execution
- test gates before merge
- diff review by human
- minimal-patch constraints
- dependency and security checks

Case study: WebArena – realistic web agents

WebArena [Zhou et al., 2024a]

- 812 tasks across e-commerce, forums, development tools, and content management
- Self-hosted web environments with functional websites
- Tasks require navigation, state tracking, reading, comparison, and action
- Original benchmark: best GPT-4-based agent at **14.41%**; humans at **78.24%**

Example WebArena-style task

“Find the cheapest product in the Electronics category that has at least a 4-star rating, and add it to my cart.”

Requires: navigation, filtering, comparison, and state-changing action.

What the gap reveals

- Fluent interaction $\neq$ reliable task completion
- Web navigation requires persistent state and error recovery
- Multi-step reasoning across pages remains fragile
- Agents must distinguish task progress from superficial page activity

WorkArena/WorkArena++ [Drouin et al., 2024b]: evaluates enterprise-style ServiceNow tasks and compositional planning.

Case study: CRMarena – professional business agents

CRMarena [[Huang et al., 2025a](#)]

- Evaluates agents on professional CRM tasks in realistic business environments
- Tests service workflows, analytics, and business reasoning
- Requires multi-step data analysis, not only web navigation
- Highlights the gap between generic tool use and domain-specific operational competence

CRMarena-Pro [[Huang et al., 2025b](#)]

- Adds multi-turn business interactions
- Evaluates more diverse professional scenarios
- Combines task completion with quality-oriented metrics

Sources: [[Huang et al., 2025a](#), [Huang et al., 2025b](#)].

Case study: ST-WebAgentBench – safety under policy

ST-WebAgentBench [Levy et al., 2024]

The **safety + trust** benchmark:

- Pairs web tasks with explicit safety/trust policies
- Introduces **Completion Under Policy (CUP)**
- Requires the agent to complete the task *and* respect policy constraints throughout

Key finding

- Completion rates **drop sharply** once policy compliance is measured
- Completing a task while violating policy is **worse than failing**
- Standard task-success metrics miss this entirely

Implication

Benchmark scores without policy-awareness can be **actively misleading** about deployment quality.

Sources: [Levy et al., 2024].

Case study: mobile and embodied agents

MobileSafetyBench [Lee et al., 2026]

- Evaluates the safety of autonomous agents that control real mobile devices
- Tasks span messaging, banking-style apps, and system settings
- Measures whether agents avoid harmful, irreversible, or unauthorized actions while completing everyday device tasks

AGENTS SAFE [Liu et al., 2025]

- Benchmarks the safety of **embodied** agents given hazardous or unsafe natural-language instructions
- Tests whether an agent recognizes danger and refuses, replans, or asks for confirmation instead of executing literally
- Extends safety evaluation from text/web agents to physically-grounded action

Sources: [Lee et al., 2026, Liu et al., 2025].

Beyond Tech: assessing agents' societal impact

What did it cost society? Whom did it mislead? How easily could it be manipulated?

DeepTRACE [Venkit et al., 2025]

- Scenario-driven robustness
- Foregrounds the **societal impact** of agent behavior, not only task completion
- Traces whether an agent's outputs are grounded, attributable, and free of fabricated or misleading evidence
- Formalizes evaluation of **groundedness and citation fidelity** as first-class metrics
- Focus on behavioral safety and trustworthiness

Agent Security Bench (ASB)

[Zhang et al., 2025]

- Formalizes **security resilience**: prompt injection, tool poisoning, adversarial observations
- Tests whether an agent's own tool outputs can be weaponized to override user or developer intent
- Complements DeepTRACE's societal lens with a security-resilience lens

Sources: [Venkit et al., 2025, Zhang et al., 2025, Narayanan Venkit et al., 2025].

Summary: Evaluation – Design Insights

1. Own your evaluation

Domain-specific and enterprise AI must own its own evaluation – you cannot let the agent grade its own homework in a domain only you understand.

2. Some guardrails are non-negotiable

Permissions, irreversible actions, and specific lookups must be enforced with 100% certainty – not left to a probabilistic grader.

3. Evaluation is part of the harness

The same tight/loose tradeoff from retrieval applies here: how strict a grader you build should match how much you trust the agent.

4. The harness must be 24/7 and proactive

Evaluation cannot be a periodic offline check – it must run continuously, surfacing issues before errors compound.

5. Objective and subjective, together

Some criteria are objective (did it pass, was permission checked); others are inherently subjective (was this a good decision). Measure both.

6. The grader must learn too

Evaluation criteria aren't fixed at launch – graders may learn by shadowing human experts, calibrating against their judgments over time.

Synthesis: retrieval and evaluation are both design surfaces

Neither retrieval nor evaluation is a one-shot, fire-and-forget step.

- Retrieval becomes iterative, adaptive, and modular precisely when the task is multi-step and open-ended – the same conditions that make outcome-only evaluation insufficient
- Behavior-centric, trajectory-level evaluation exists precisely because task success alone hides how an agent got there
- Safety and sociotechnical benchmarks (Agent-SafetyBench, ST-WebAgentBench, DeepTRACE, ASB, MobileSafetyBench, AGENTSAFE) generalize across domains – web, mobile, embodied, and scientific
- Trajectories are not just execution logs to discard after scoring: they are raw material for both evaluation and self-improvement

Carry this forward

Use the retrieval taxonomy, the trajectory-level evaluation checklist, the safety-risk table, and the evaluation ladder from this block as templates.

Block 3: Agents in Finance

- From general-purpose evaluation to a domain where an agent's mistake is not a bad demo – it is a bad trade
- The planner–executor–verifier architecture, applied to earnings analysis, portfolio allocation, and risk assessment
- FinAgent, FinRobot, and FinMem, evaluated with the same checklist just introduced

Presented by **Enrico Santus** (Bloomberg), returning after this block.

Sources: Systems covered in Block 3: [\[Zhang et al., 2024a, Zhou et al., 2024b, Yu et al., 2023\]](#).

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